CREEP AND SHRINKAGE IN CONCRETE STRUCTURES

INTRODUCTION

Definitions

- Creep is time dependent deformations of concrete under permanent loads (self weight), PT forces and permanent displacement.
- INCREASE IN STRAIN UNDER SUISTAINED STRESS
- When concrete is subjected to compressive loading it deforms instantaneously. This immediate deformation is called instantaneous strain.
- Now, if the load is maintained for a considerable period of time, concrete undergoes additional deformations even without any increase in the load. This time-dependent strain is termed as creep.

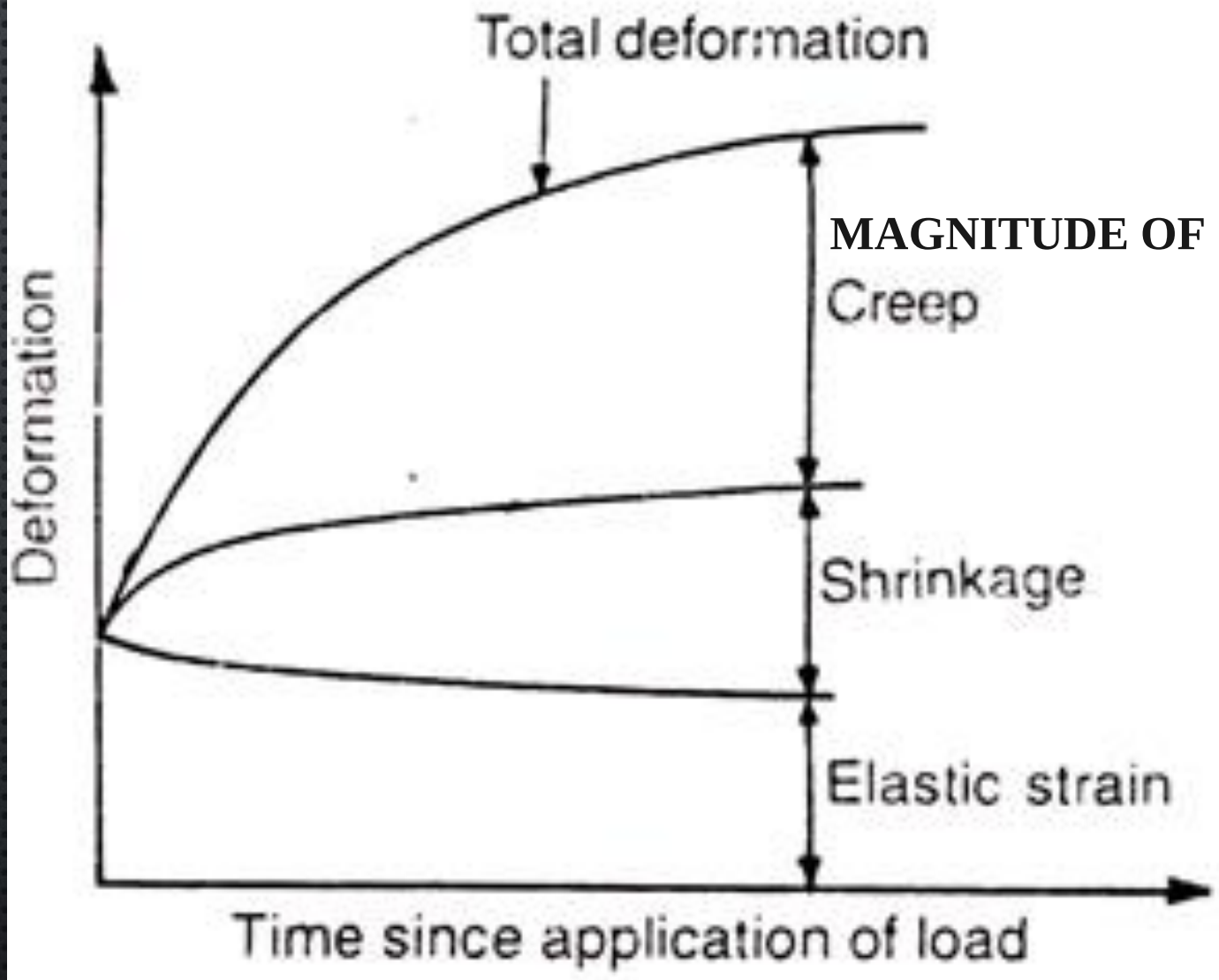


Fig. 15.20. Time dependent deformation in concrete subjected to a sustained load

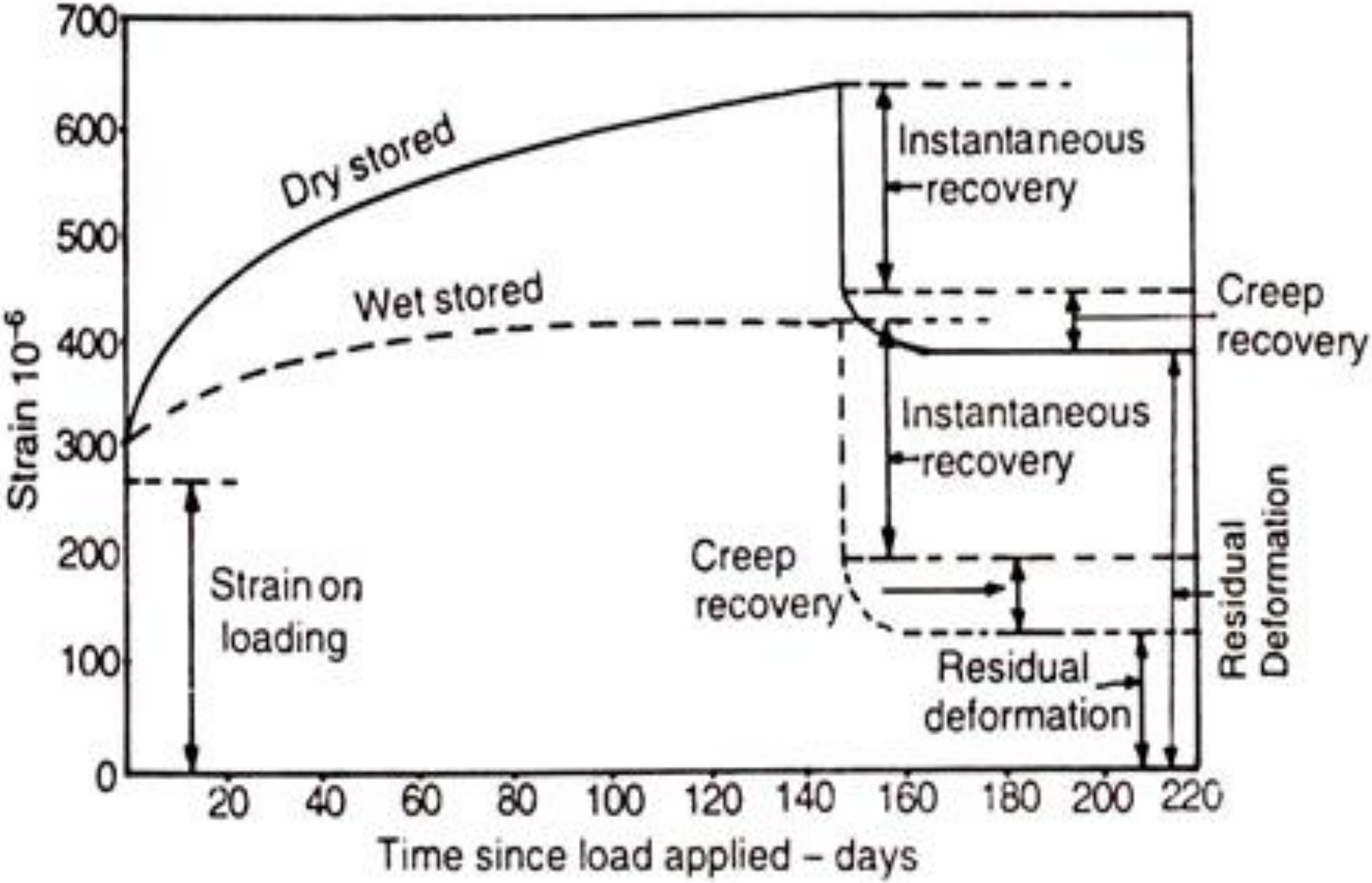


Fig. 15.7. Creep and creep recovery of concrete stored in water and air from the age of 28 days

CREEP





CREEP:





CREEP:



Factors Affecting Creep

- Concrete mix proportion
- Aggregate properties
- Age at loading
- Curing conditions
- Cement properties
- Temperature
- Stress level

FACTORS AFFECTING CREEP

1..Concrete mix proportion:

- The amount of paste content and its quality is one of the most important factors influencing creep.
- A poorer paste structure undergoes higher creep.
- creep increases with increase in water/cement ratio.

1.Concrete mix proportion:

- creep is inversely proportional to the strength of concrete.
- All other factors which are affecting the water/cement ratio are also affecting the creep.



2..Aggregate properties:

- Aggregate undergoes very little creep.
- It is really the paste which is responsible for the creep.
- Aggregates influence creep of concrete through a restraining effect on the magnitude of creep.
  - HIGHER MODULUS OF ELASTICITY LESSER WILL BE CREEP
  - LIGHT WEIGHT AGGREGATE SHOW HIGHER CREEP THAN NORMAL WEIGHT AGGREGATE

FACTORS AFFECTING CREEP

2..Aggregate properties:

Fine aggregates

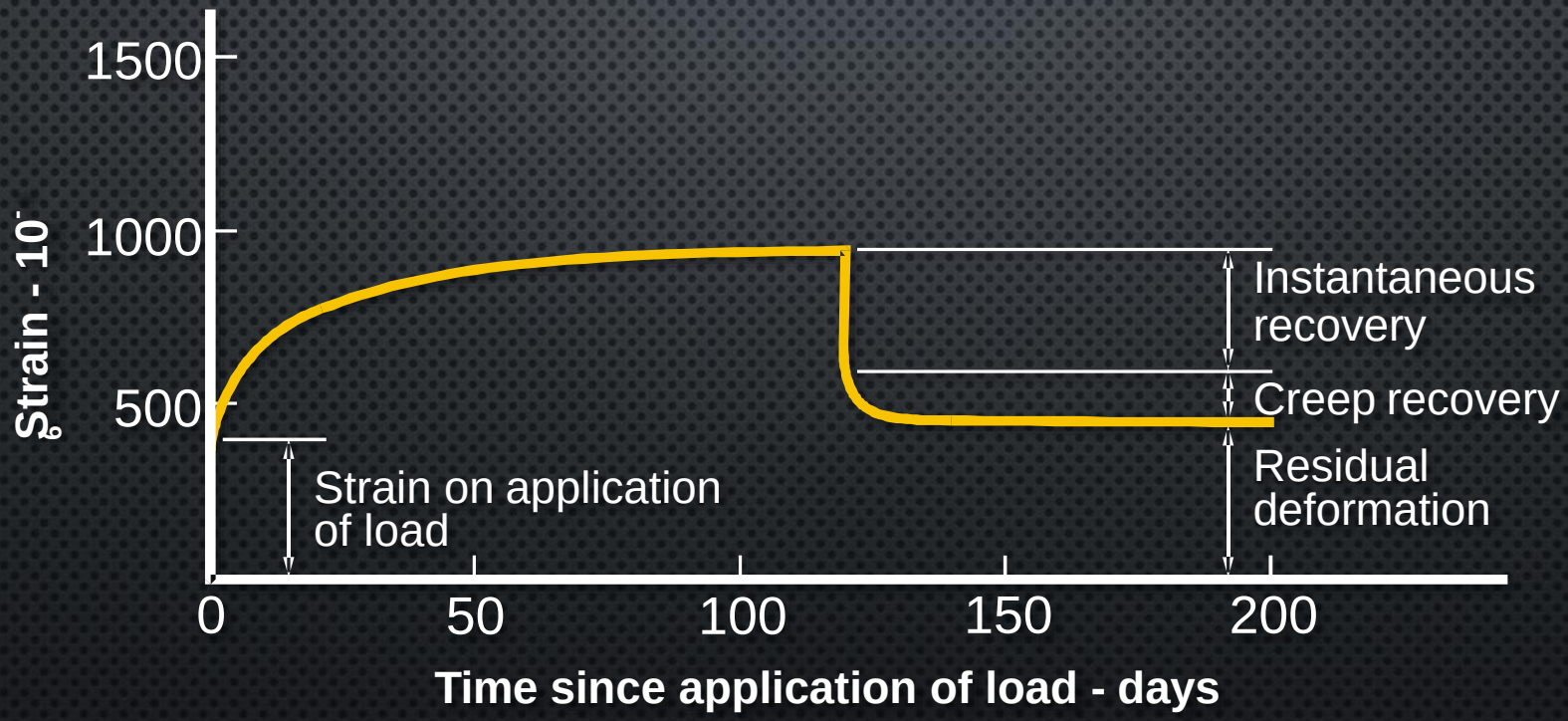


Coarse aggregates



- 3.Age at loading
- Age at which a concrete member is loaded will have a predominant effect on the magnitude of creep.
- The quality of gel improves with time. Such gel creeps less.
- Whereas a young gel under load being not so stronger creeps more.
- The moisture content of the concrete being different at different age also influences the magnitude of creep.

FACTORS AFFECTING CREEP



4.Curing condition:

- In view of the smallness of creep strains, the amount of water expelled during creep from the micro pores into the macro pores (or vice versa) must also be small, probably much less than 0.1 percent of the volume of concrete (since typically creep strains do not exceed 0.001, and even this is not due entirely to water but also to expelled solids).

- Larger the curing smaller the creep.

FACTORS AFFECTING CREEP

4..Curing condition:



5.Cement properties:

- The type of cement effects creep in so far as it influence the strength of the concrete at the time of application of load.
- Fineness of cement affects the strength development at early ages and thus influence creep.
- The finer the cement the higher its gypsum requirement so that re grinding of cement in laboratory without the addition of gypsum produces an improperly retarded cement, which exhibits high creep.

FACTORS AFFECTING CREEP

6..Temperature:

- The rate of creep increases with temperature up to about 70^0 C when, for a 1:7 mix and 0.6 w/c ratio.
- It is approximately 3.5 times higher than at 21^0 C.
- Between 70^0 C and 96^0 C it drops off to 1.7 times tan at 21^0 C.

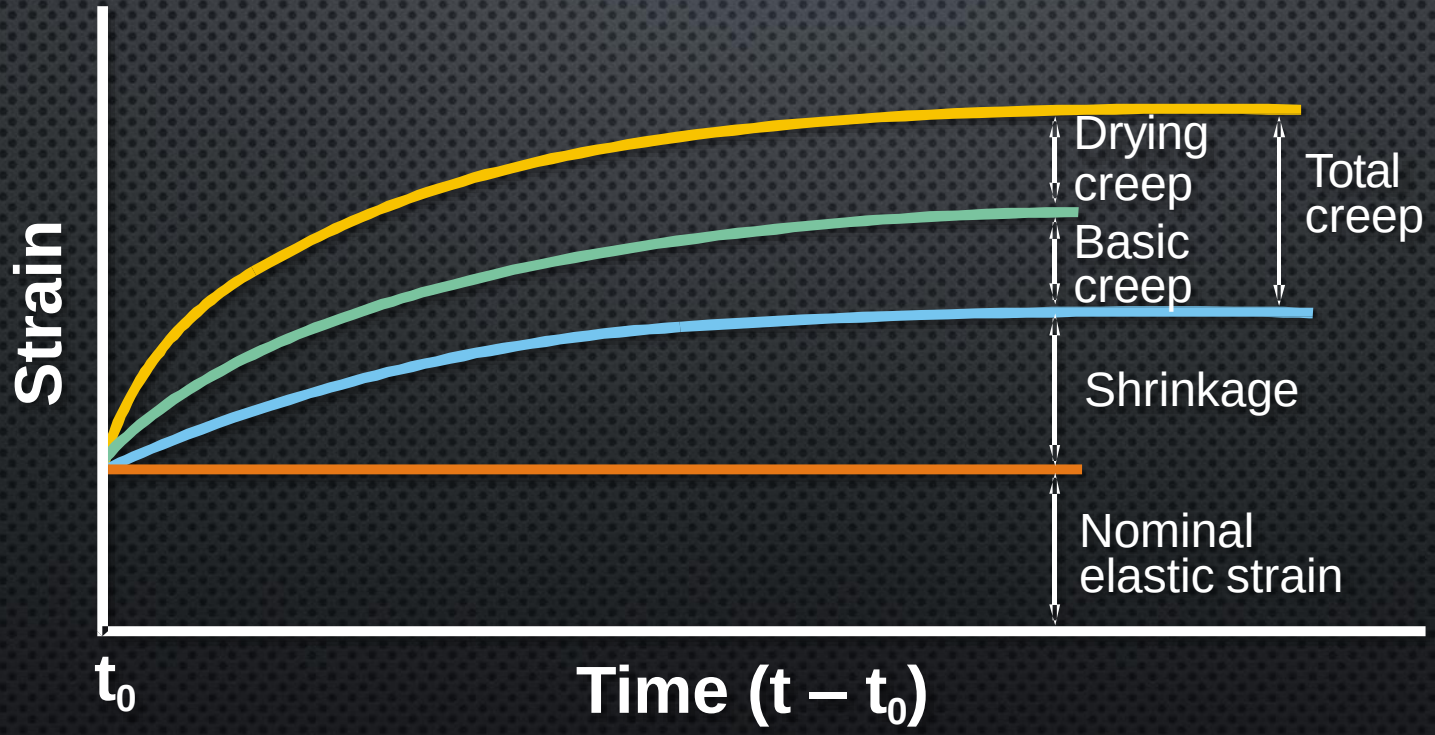
6..Temperature:

- As far as low temperature is concerned, freezing produces a higher initial rate of creep but it quickly drops to zero.
- At temperature between 100C and 300C, Creep is about one half of creep at 210C.

7..Stress level:

- There is a direct proportion between creep and applied stress.
- There is no lower limits of proportionality because concrete undergoes creep even at very low stress.
- Higher the stress higher will be the creep.

CREEP ANALYSIS



CREEP ANALYSIS

Relationship between creep and elastic deformations

$$\varepsilon_{cr} = \varepsilon_{el}\,\phi = \frac{\sigma}{E_{28}}\,\phi$$

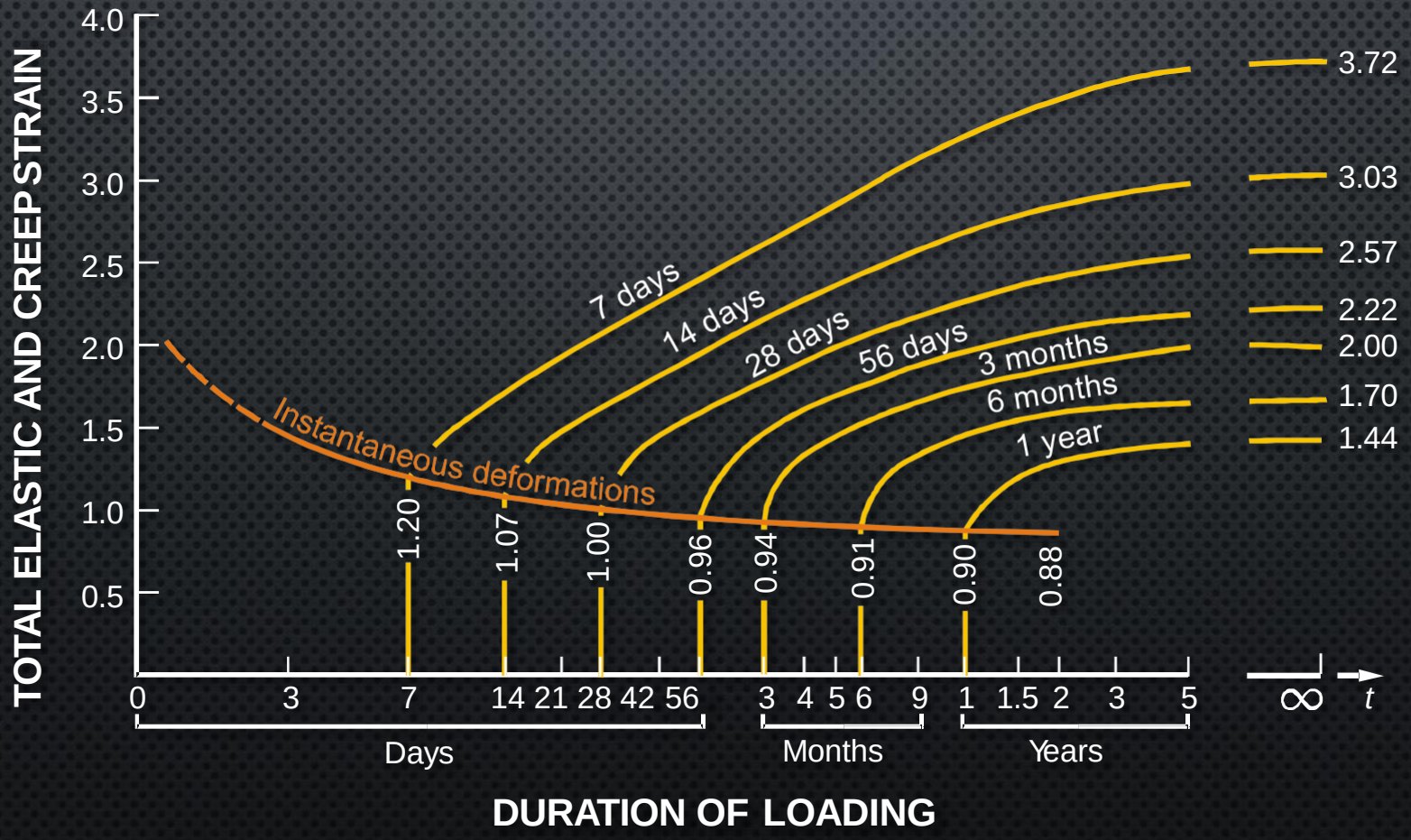
where: ε_{cr} = creep strain

ε_{el} = elastic strain

σ = stress

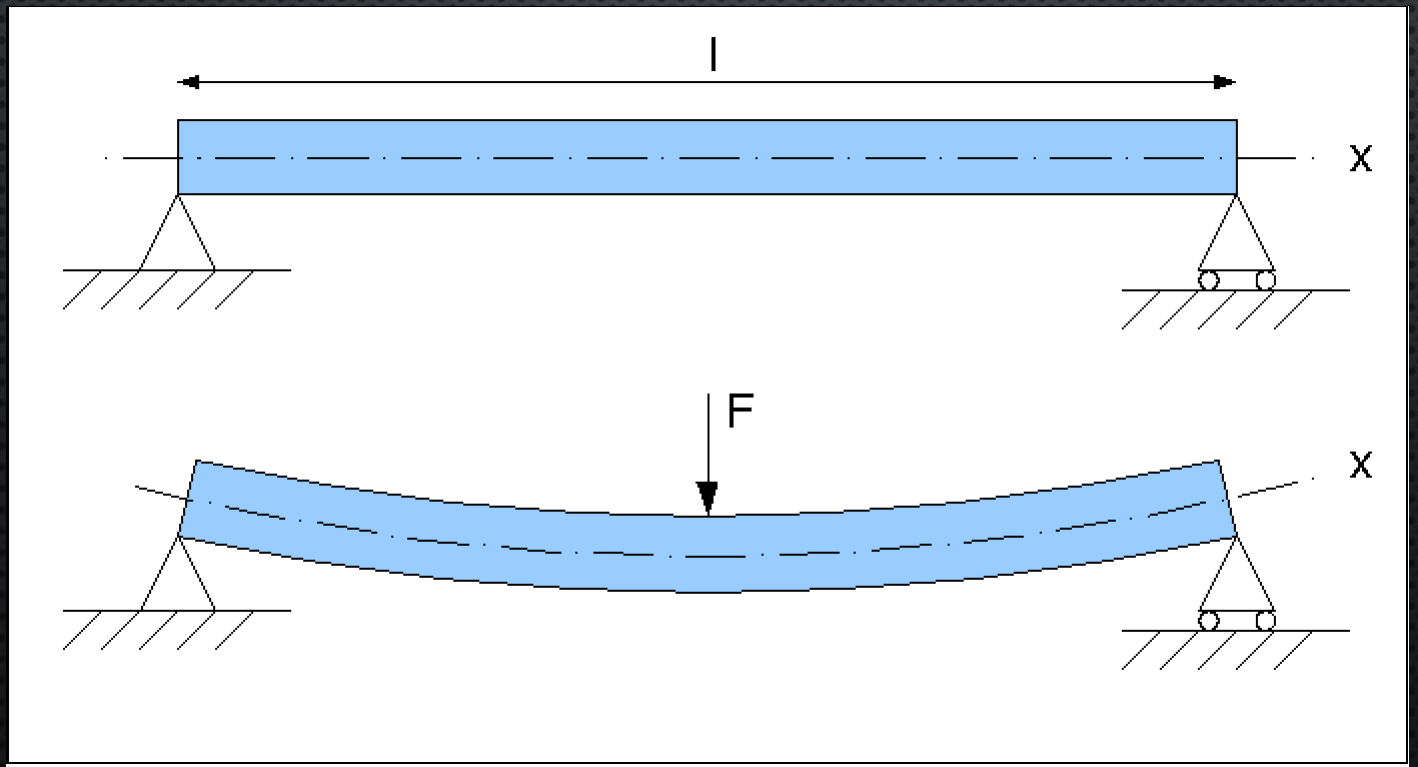
E_{28} = elastic modulus of concrete at age 28 days

ϕ = creep factor


TOTAL ELASTIC AND CREEP STRAIN
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
7 days
14 days
28 days
56 days
3 months
6 months
1 year
Instantaneous deformations
1.20
1.07
1.00
0.96
0.94
0.91
0.90
0.88
3.72
3.03
2.57
2.22
2.00
1.70
1.44
0 3 7 14 21 28 42 56 3 4 5 6 9 1 1.5 2 3 5
Days
Months
Years
∞L
t
DURATION OF LOADING

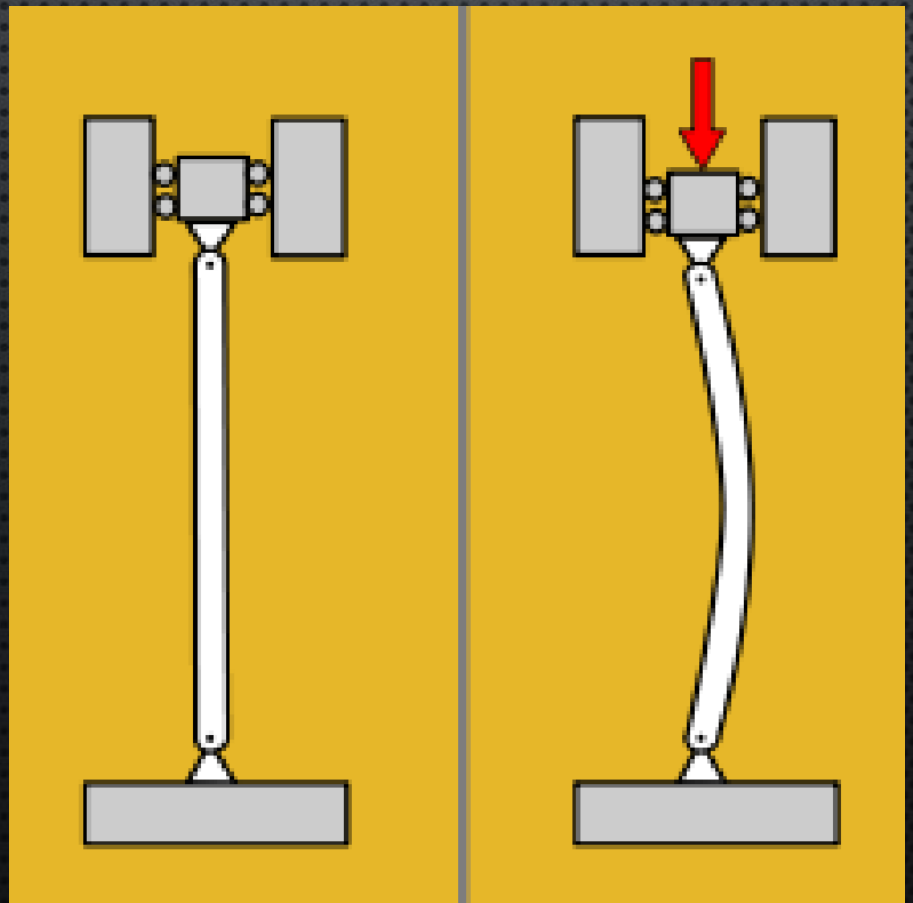
EFFECTS OF CREEP ON CONCRETE STRUCTURES

- **In reinforced concrete beams, creep increases the deflection with time and may be a critical consideration in design.**



EFFECTS OF CREEP ON CONCRETE STRUCTURES

- **In eccentrically loaded columns, creep increases the deflection and can load to buckling.**



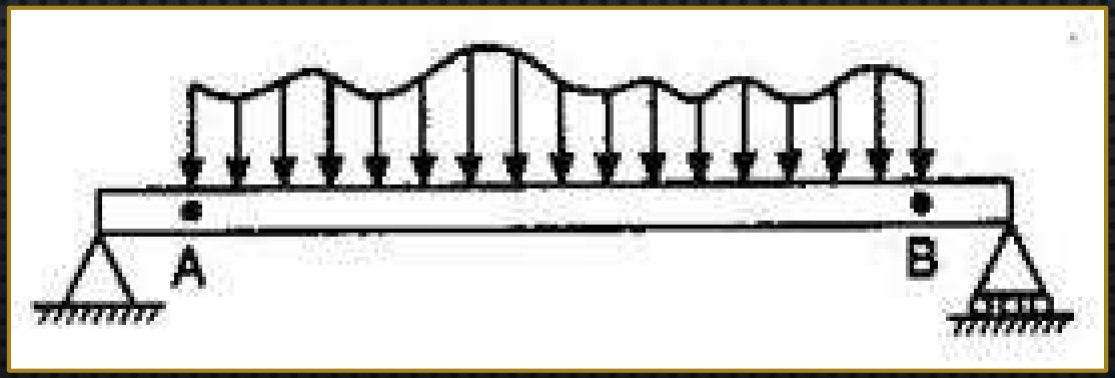
EFFECTS OF CREEP ON CONCRETE STRUCTURES

- **Loss of pre stress due to creep of concrete in pre stressed concrete structure.**



EFFECTS OF CREEP ON CONCRETE STRUCTURES

- Creep property of concrete will be useful in all concrete structures to reduce the internal stresses due to non-uniform load or restrained shrinkage.



EFFECTS OF CREEP ON CONCRETE STRUCTURES

In mass concrete structures such as dams, on account of differential temperature conditions at the interior and surface, creep is harmful and by itself may be a cause of cracking in the interior of dams



INTRODUCTION:

Definition:

Volume change is one of the objectionable property of concrete which affect long term strength and durability of concrete.

- Shrinkage is shortening of concrete due to drying and is independent of applied loads.
- Shrinkage of concrete is the time-dependent strain measured in an unloaded and unrestrained specimen at constant temperature.

SHRINKAGE:



SHRINKAGE:





Factors Affecting Shrinkage

- Drying conditions
- Time
- Water cement ratio
- Type of Aggregate
- Type of Cement
- Size and shape of Specimens

FACTORS AFFECTING SHRINKAGE

1..Drying conditions:

- The most important factor is the drying condition or the humidity in the atmosphere.
- No shrinkage will occur if the concrete is placed in one hundred percent relative humidity.



FACTORS AFFECTING SHRINKAGE

2. Time:

- The shrinkage rate will decrease rapidly with time.
- It has been documented that fourteen to thirty-four percent of the twenty year shrinkage will occur within two weeks of it being poured.
- Within one year of the concrete being poured, shrinkage will be about sixty-six to eighty-five percent of the twenty year shrinkage.

FACTORS AFFECTING SHRINKAGE

3..Water cement ratio:

- The water to **cement ratio** will influence the amount of shrinkage that occurs.
- The concrete's richness also affects the shrinkage.
- The process of swelling and then drying affects the concrete's integrity and the shrinkage.

FACTORS AFFECTING SHRINKAGE

4.Type of Aggregate:

- Aggregates with low modulus of elasticity causes large shrinkage of concrete
- Increase in maximum size causes restraining effect hence decrease shrinkage.

FACTORS AFFECTING SHRINKAGE

5.Type of Cement:

- Shrinkage is somewhat more in rapid hardening cement than other types of cement.

FACTORS AFFECTING SHRINKAGE

6.Size and Shape of Specimen:

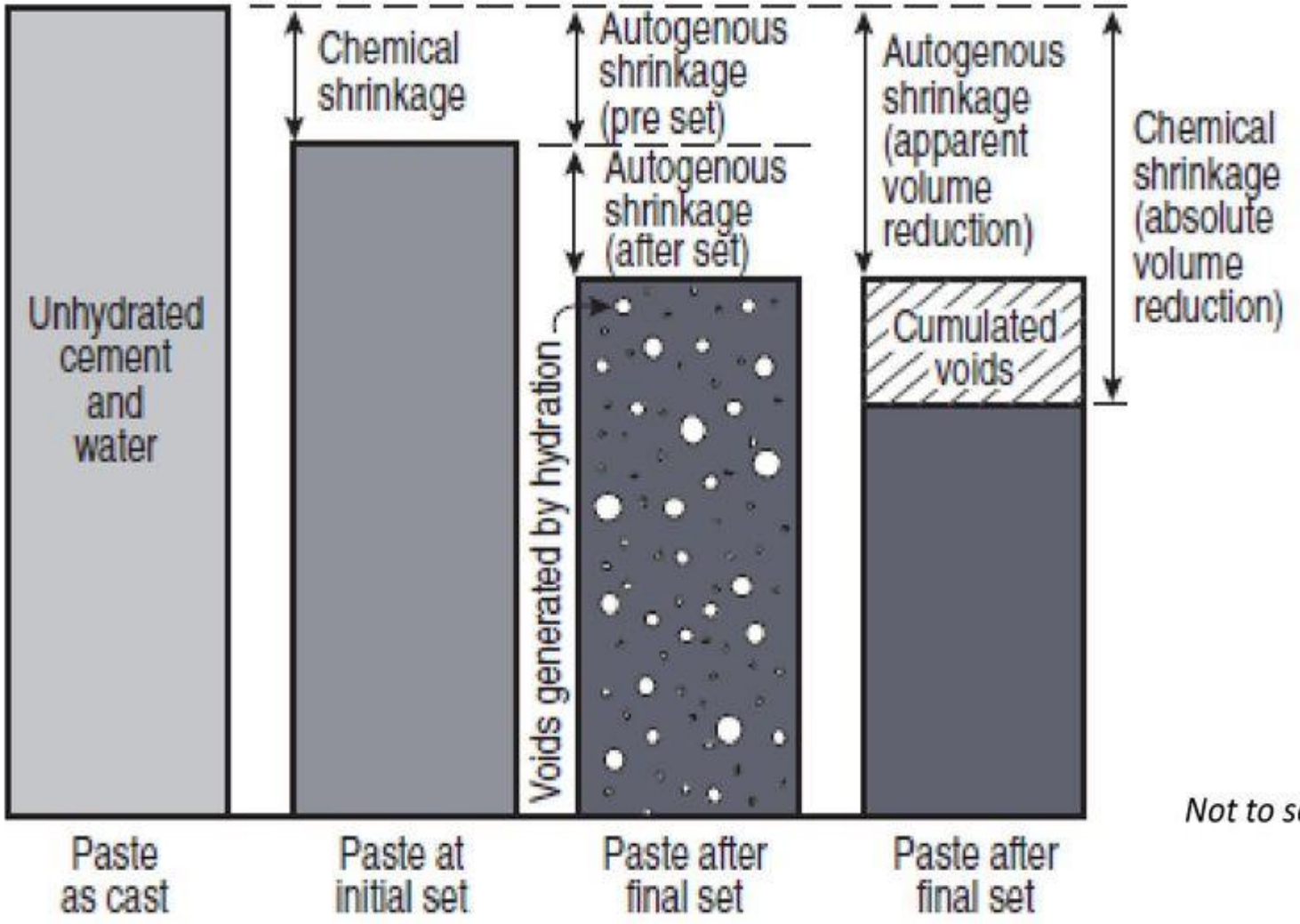
- Shrinkage decreases with increase in size of specimen.

INTRODUCTION:

Types of shrinkage

- Plastic Shrinkage
- Drying Shrinkage
- Autogeneous Shrinkage
- Carbonation Shrinkage

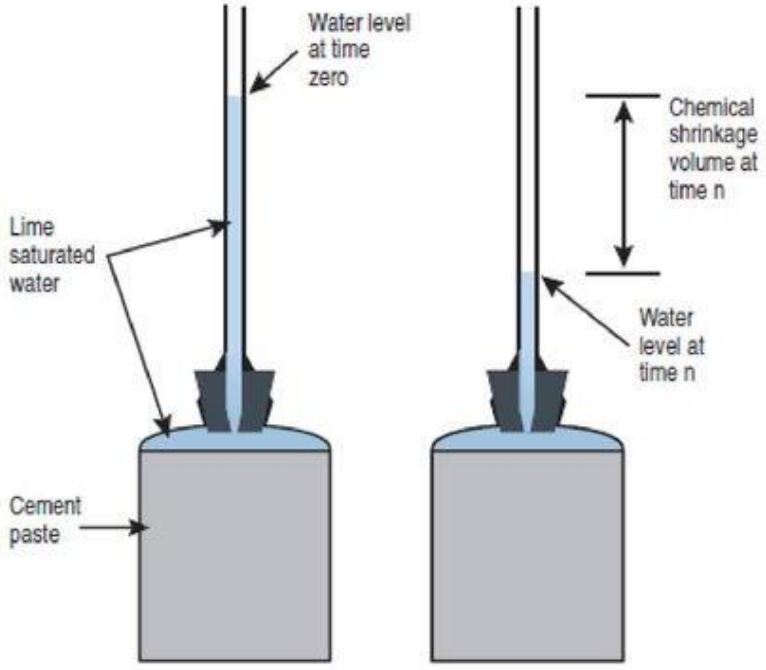
Early Age Volume Changes

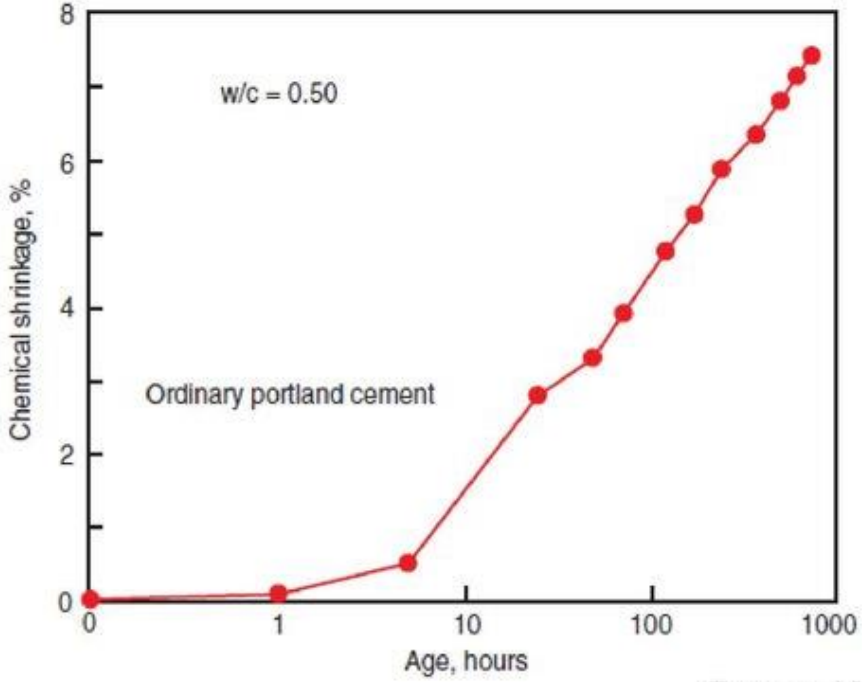


Not to scale.

Chemical Shrinkage

ASTM C1608- *Standard Test Method for Chemical Shrinkage of Hydraulic Cement Paste*





Tazawa 1999

- volume of the hydration product is smaller than volume of un-hydrated cement and water (0.06Ch) & is chemical shrinkage

1..Carbonation shrinkage:

- Carbonation shrinkage is a phenomenon very recently recognized and is very important.
- Carbon dioxide present in the atmosphere reacts in the presence of water with hydrated cement.
- Calcium hydroxide gets converted to calcium carbonate and also some other cement compounds are decomposed.
- Such a complete decomposition of calcium compound in hydrated cement is chemically possible even at the low pressure of carbon dioxide in normal atmosphere.
- Carbonation penetrates beyond the exposed surface of concrete only very slowly.

1..Carbonation shrinkage:

- The rate of penetration of carbon dioxide depends also on the moisture content of the concrete and the relative humidity of the ambient medium.
- Carbonation is accompanied by an increase inweight of the concrete and by shrinkage.
- Carbonation shrinkage is probably caused by the dissolution of crystals of calcium hydroxide and deposition of calcium carbonate in its place.
- As the new product is less in volume than the product replaced, shrinkage takes place.

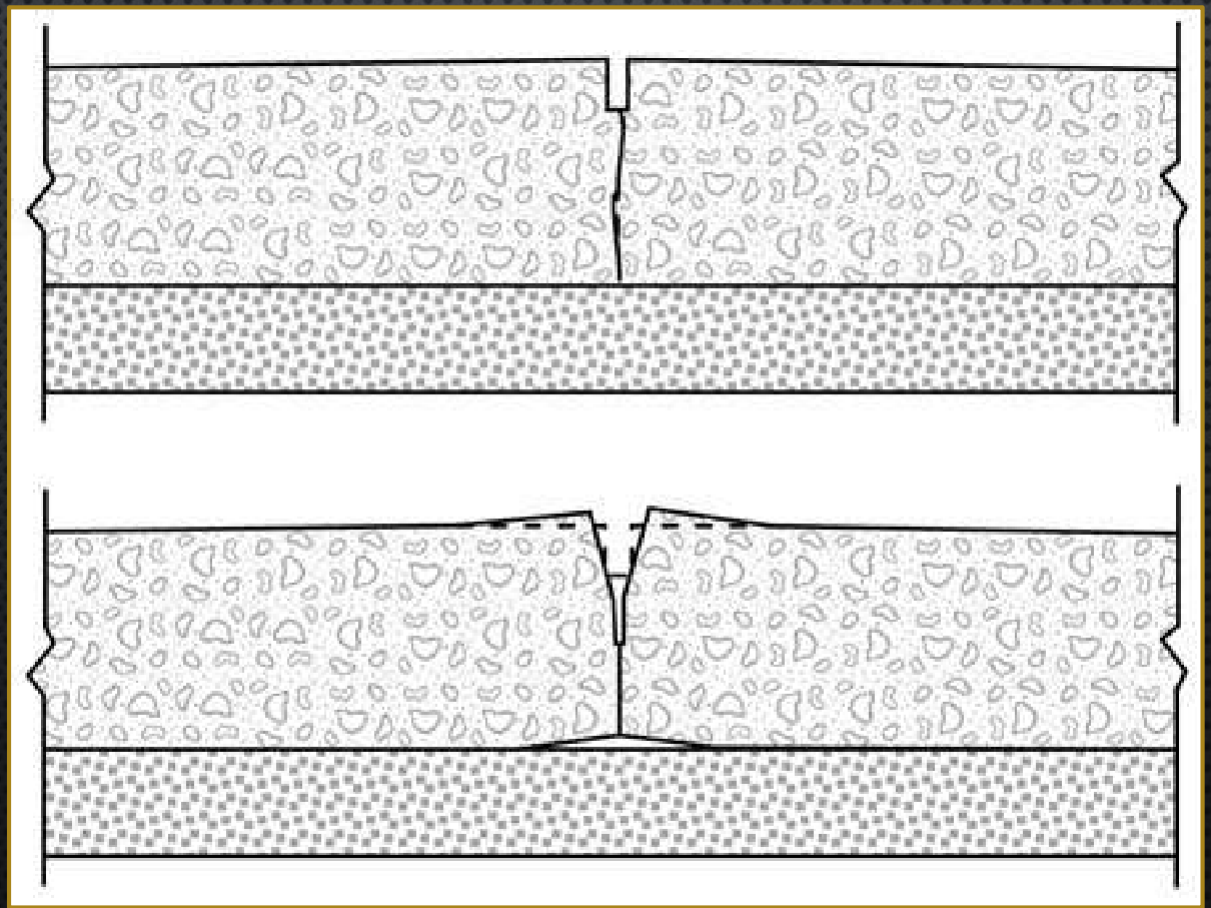
TYPES OF SHRINKAGE

1..Carbonation shrinkage:



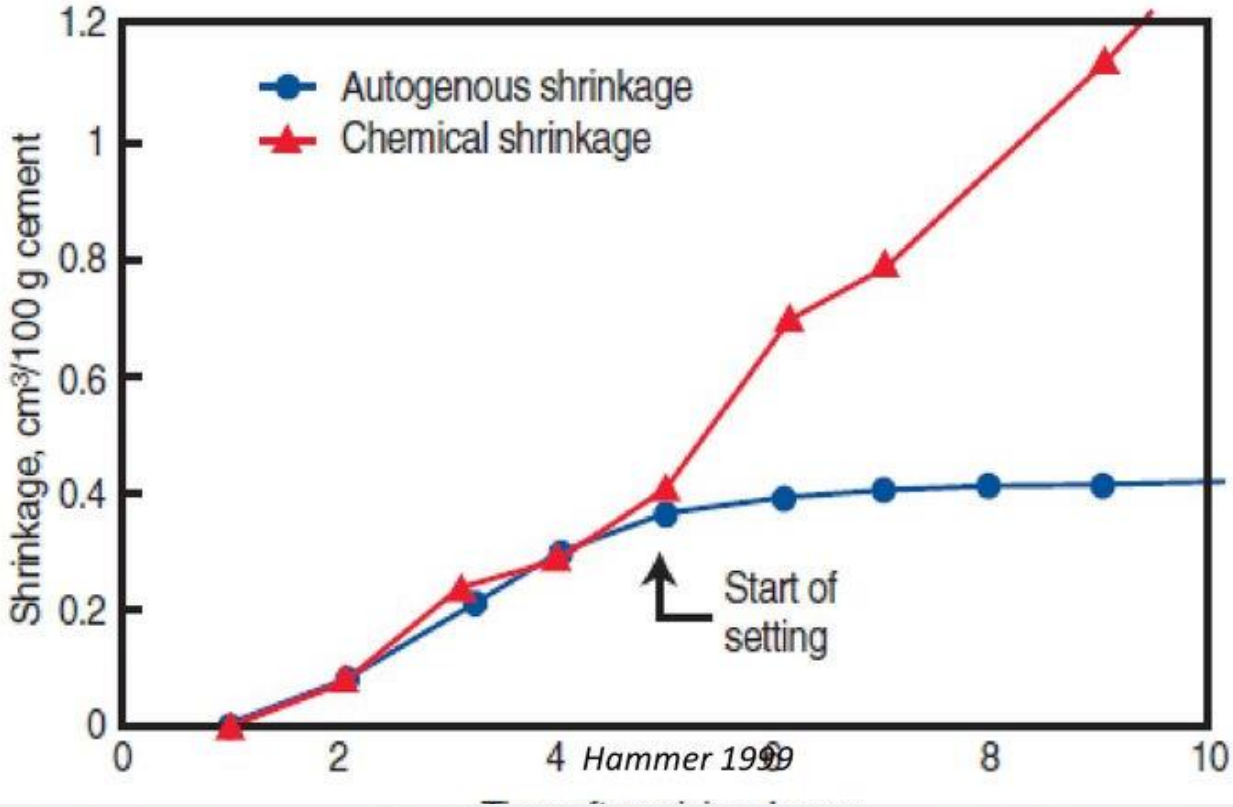
TYPES OF SHRINKAGE

1..Carbonation shrinkage:



Autogenous Shrinkage

ASTM C1698- *Standard Test Method for Autogenous Strain of Cement Paste and Mortar*



Autogenous shrinkage is a macroscopic reduction in length under constant temperature and without any moisture migration to or from the concrete.

2.Autogeneous shrinkage:

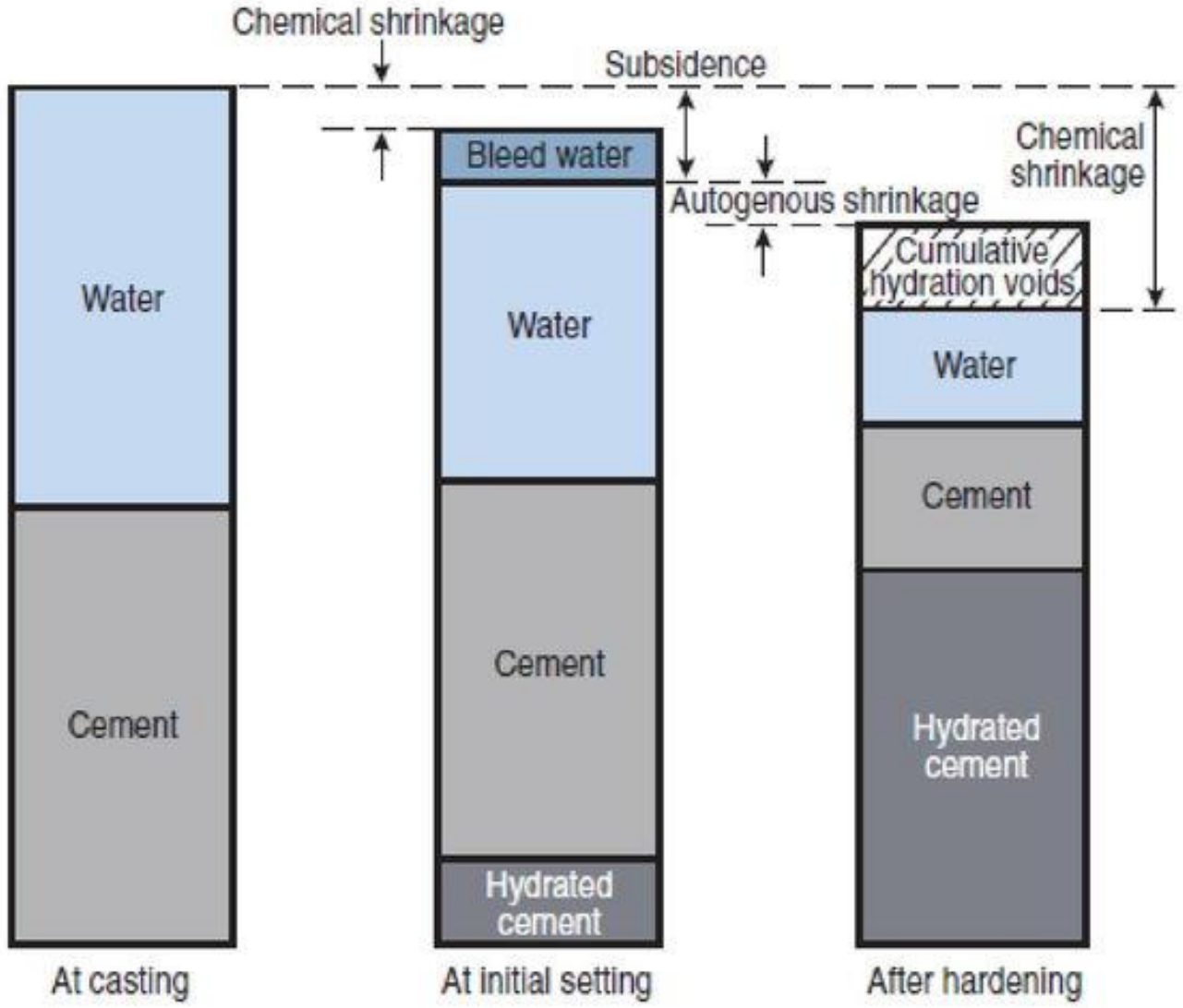
- In a conservative system i.e. where no moisture movement to or from the paste is permitted, when temperature is constant some shrinkage may occur. The shrinkage of such a conservative system is known as autogeneous shrinkage.
- Autogeneous shrinkage is of minor importance and is not applicable in practice to many situations except that of mass of concrete in the interior of a concrete dam.

TYPES OF SHRINKAGE

2..Autogeneous shrinkage:



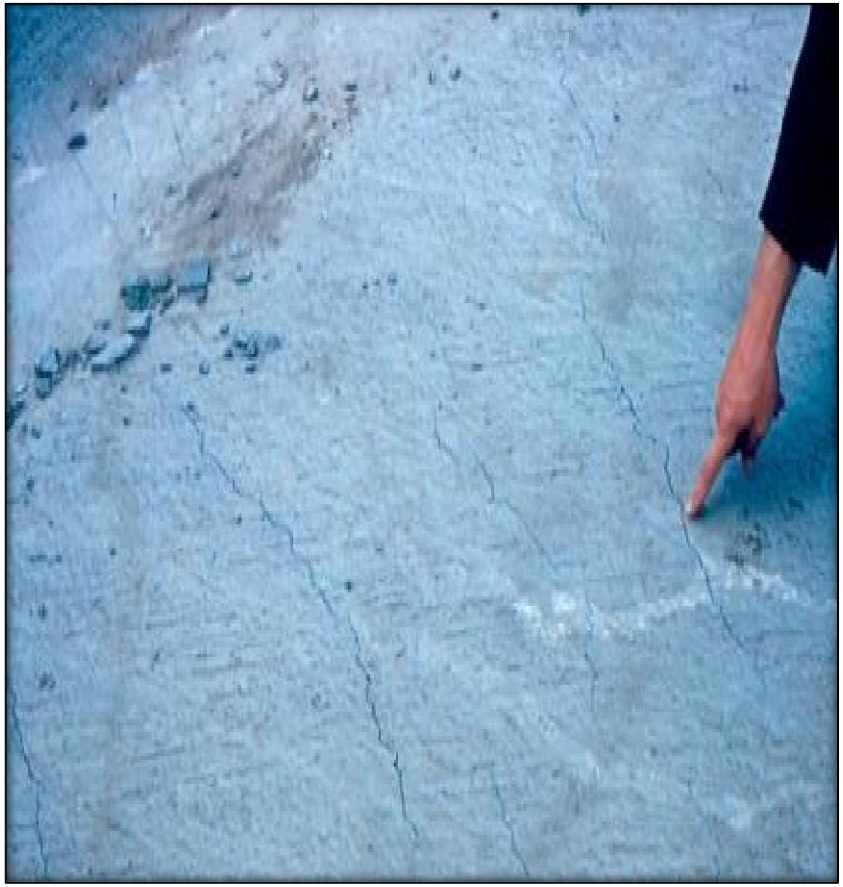
Autogenous Shrinkage

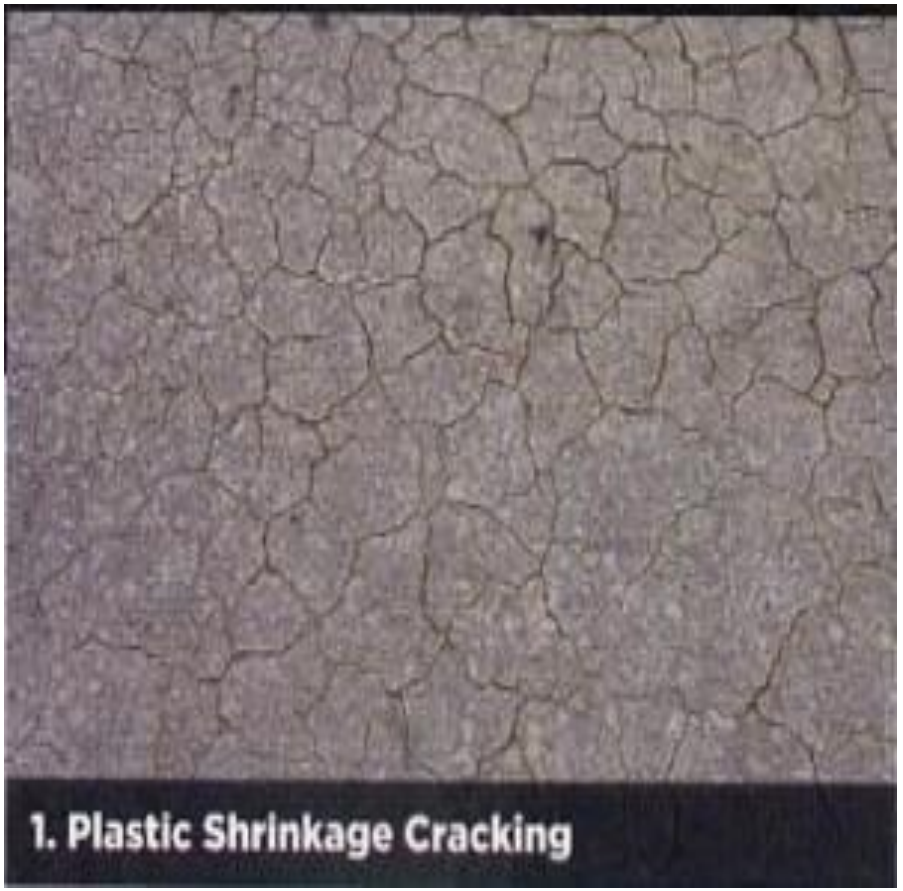


3.Plastic Shrinkage

- Shrinkage which takes place before concrete has set.
- Occurs during the first few hours after fresh concrete is placed.
- During this period, moisture may evaporate faster from the concrete surface than it is replaced by bleed water from lower layers of the concrete mass.

Plastic cracking (Plastic Shrinkage Cracking) is cracking that occurs in the surface of the fresh concrete soon after it is placed and while it is still plastic.





TYPES OF SHRINKAGE

3..Plastic shrinkage:

- Plastic shrinkage happens soon after the concrete is poured in the forms.
- The water evaporates and results in a reduction of volume, this causes the concrete on the surface to collapse.
- The aggregate particles or the reinforcement comes in the way of subsidence due to which cracks may appear at the surface or internally around the aggregate or reinforcement

3..Plastic shrinkage:

- High water/cement ratio, badly proportioned concrete, rapid drying, greater bleeding, unintended vibration etc., are some of the reasons for plastic shrinkage.
- Plastic shrinkage can be reduced mainly by preventing the rapid loss of water from surface.
- It can be reduced by covering the surface with polyethylene sheeting immediately after it is poured.

TYPES OF SHRINKAGE

3..Plastic shrinkage:



TYPES OF SHRINKAGE

3..Plastic shrinkage



TYPES OF SHRINKAGE

3..Plastic shrinkage



4) Drying Shrinkage

- When a hardened concrete, cured in water, is allowed to dry it first loses water is drawn out of its cement gel.

- After an initial high rate of drying shrinkage concrete continues to shrink for a long period of time but at a continuously decreasing rate.

4..Drying shrinkage:

- Just as the hydration of cement is an ever lasting process, the drying shrinkage is also an ever lasting process when concrete is subjected to drying conditions.
- The loss of free water contained in hardened concrete, does not result in any appreciable dimension change.
- It is the loss of water held in gel pores that causes the change in the volume

4..Drying shrinkage:

- Under drying conditions, the gel water is lost progressively over a long time, as long as the concrete is kept in drying conditions.
- The magnitude of drying shrinkage is also a function of the fineness of gel.
- The finer the gel the more is the shrinkage.
- It has been pointed out earlier that the high pressure steam cured concrete with low specific surface of gel, shrinks much less than that of normally cured cement gel.

TYPES OF SHRINKAGE

4..Drying shrinkage:



TYPES OF SHRINKAGE

4..Drying shrinkage:



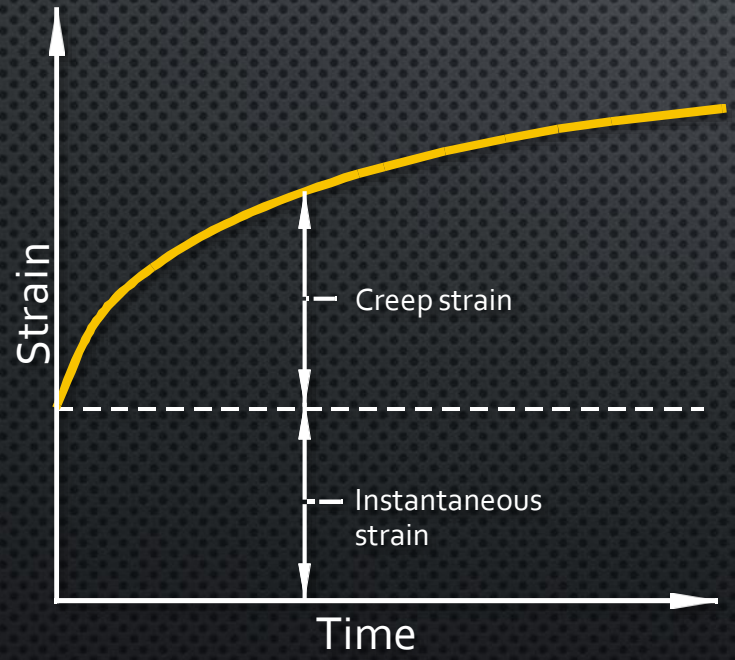


TYPES OF SHRINKAGE

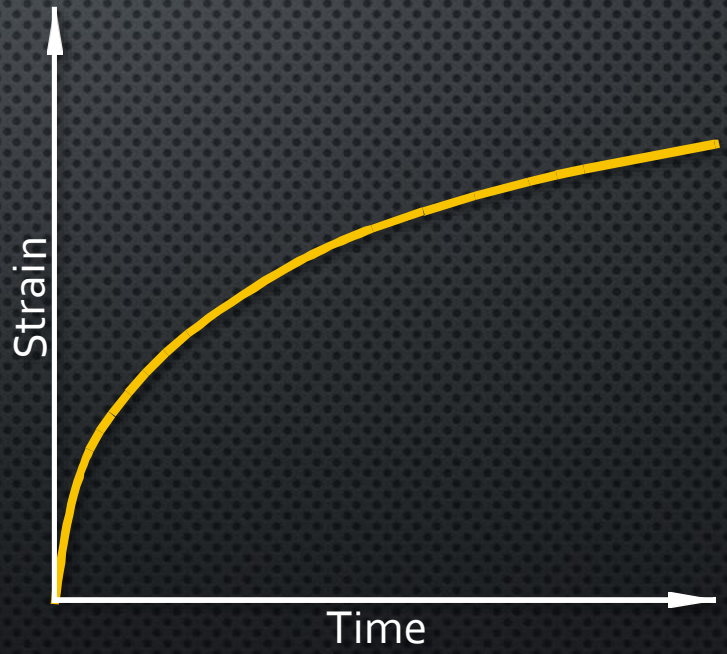
4..Drying shrinkage:



CREEP AND SHRINKAGE TYPICAL TIMECURVE



TYPICAL CREEP –TIMECURVE



TYPICAL SHRINKAGE –TIMECURVE

EFFECTS OF SHRINKAGE

- **Shrinkage of concrete between movement joints causes joints to open or makes it wider. Therefore joints must be designed to accommodate the widening caused by shrinkage.**



EFFECTS OF SHRINKAGE

- **Where other materials, such as ceramic tiles, are fixed on top of concrete surface, shrinkage of the concrete causes relative movement between the different materials. The resulting stresses can cause failure at the interface.**



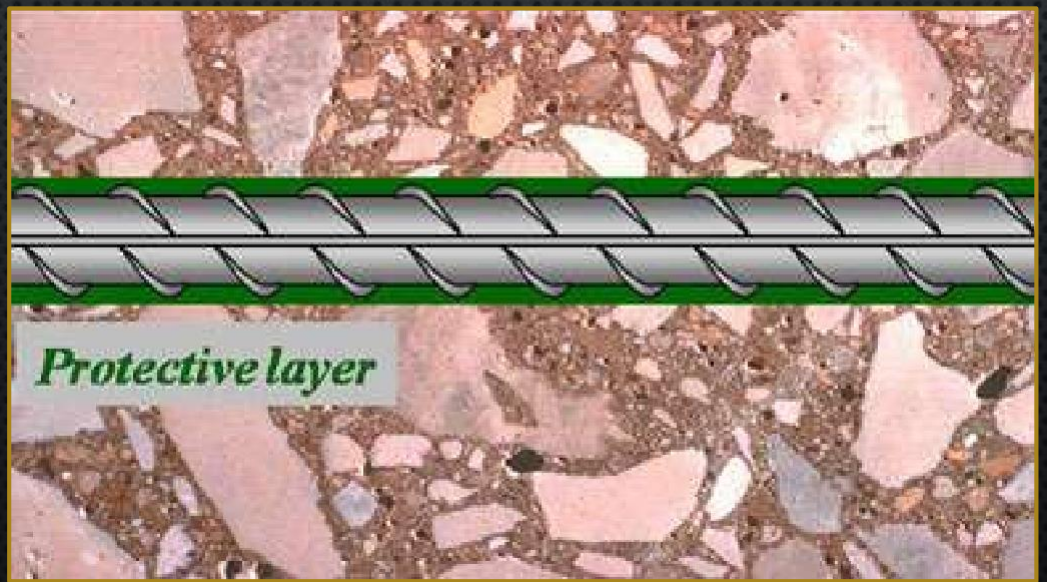
EFFECTS OF SHRINKAGE

- **If shrinkage is restrained, the concrete is put into tension and when tensile stress becomes equal to tensile strength, the concrete cracks.**



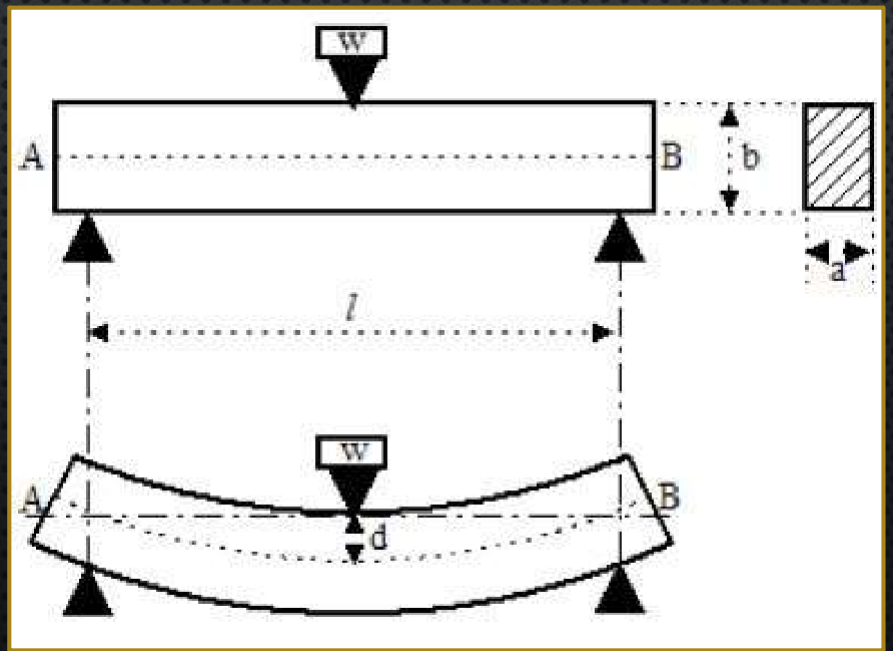
EFFECTS OF SHRINKAGE

- **Shrinkage of the concrete causes the concrete to grip reinforcing bars more tightly. This increases friction between concrete and steel and so improves bond strength, especially for plain bars**



EFFECTS OF SHRINKAGE

- **The deflection of flexural members is increased by shrinkage. This is because the lightly reinforced compression zone is free to shrink more than heavily reinforced tension zone.**



EFFECTS OF SHRINKAGE

- **Shrinkage causes a reduction in pre-stressing force. When calculating pre-stressing forces, designers take into account to ensure that residual stress is structurally adequate.**

PREVENTION OF SHRINKAGE

- Provide shun shades in case of slab construction to control the surface temperature.



PREVENTION OF SHRINKAGE

- **Dampen the subgrade of concrete before placement it is liable to water absorption but should not over damp.**



PREVENTION OF SHRINKAGE

- **Try to start the curing soon after finishing**



PREVENTION OF SHRINKAGE

- Use chemical admixtures to accelerate the setting time of concrete.

In order to avoid the negative impacts of long-term creep and shrinkage:

1. Good understanding of creep and shrinkage behaviors.
2. Accurate estimation of creep and shrinkage on structural concrete design.
3. Proper counter measures of long-term creep and shrinkage effects.
4. Implement simple structural details .



THANK YOU

ANY QUESTIONS?